

Reverberations

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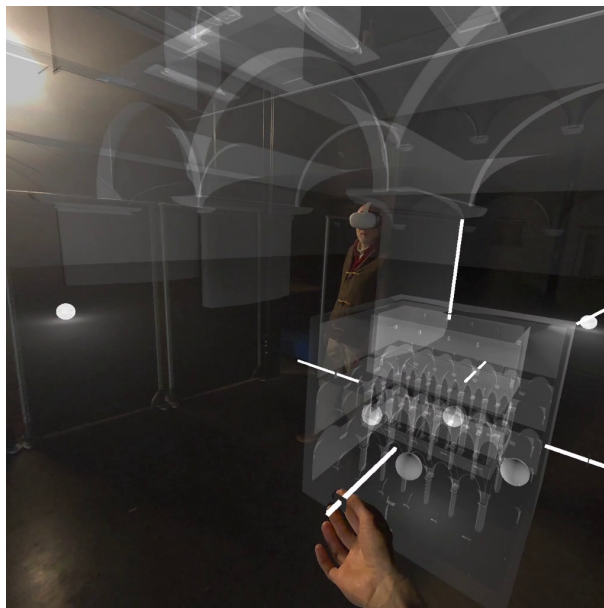


Figure 1: View of the installation as seen through the mixed-reality headset by one of the participants

ABSTRACT

Reverberations is a mixed-reality artistic installation in which participants explore a virtual building placed in the physical space. Sound sources placed in this building are spatialised, so that participants can listen to recordings of speech and instruments, which evolve in legibility as the participants move around. The building also stretches dynamically, altering the perception of the sounds.

1 PROJECT URL

<https://beaumontsebastien1.wixsite.com/website/single-project-1>

2 DESCRIPTION OF ARTWORK

“Reverberations” was born of a childhood experience. At the age of 5, I entered a church during a service. As I crossed the threshold of the building, I was greeted by an enveloping reverberation of murmurs that masked the meaning, but also let out a soft, reassuring depth. The closer I got to the source, the more I gained access to the meaning, and the more I lost the initial poetry that had given me the pleasure of the moment. I stepped back, lost the meaning and regained the pleasure. I would later learn that the tipping point with which I had been “playing” is located at a given relative distance, known as the “critical distance”, between the free field (or direct signal from the sound source) and the diffuse field (reverberated signal). This acoustic experience led me much later to a thought that would prove essential and existential for me: quite simply, to

question my “position” in the world around me, as a geographical position of course, but also as political (in the sense of Polis) and spiritual one. How much “reality” do I want and can I bear to make life acceptable? Is this the way to enjoy life? How do we find the balance between reality and blur? How much blur do we inject into reality to survive it, according to our singular psycho-affective and psychomotor conditions?

To answer all these questions, we set out to invent a manageable tool, offering the empirical trial-and-error possibilities needed to find and experience the critical distance. This is where virtuality comes in.

In “Reverberations”, two participants each find their own critical distance by moving, using mixed-reality headsets, through a virtual building with multiple sound sources symbolised by spheres. What participants perceive of these sound sources will depend on their spatial relation to them (distance, angle) but also on the geometry of the building, which can be modified dynamically. These aspects will affect the spatialisation, reverberation and filter applied to the sources, radically changing their legibility. Walking through this building that will stretch and move, participants will therefore each experience the critical distance of spoken texts and instrumental phrases.

“Reverberations” consists of two modes, which alternate during the conference.

In the interactive installation mode, which loops over 5 minutes, the two participants will experience several phases, with the apparition and changes of the building and the apparition of multiple sound sources, while freely exploring the environment. They are eventually given a world-in-miniature control allowing them to transform the building themselves.

In the performance mode, a performer uses the world-in-miniature control to shape the environment to their own will, forcing participants to reconsider their relationship with space and

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sound.

3 ARTIST INFORMATION

Sebastien Beaumont had his first guitar at the age of 13. After ensemble classes, he then followed one year of instrument practice at the CIM of Paris, and 4 years at the Lille Conservatory. He then met most of the musicians of what was going to become CIRCUM, an then Muzzix. Many projects followed crossing roads with composers such as Tom Johnson, Michael Pisaro, Satoko Fuji or Antony Pateras. He was then a permanent member of the Muzzix big orchestra under the direction of Christian Pruvost. He collaborated with the Dédalus ensemble on music by the American composer Moondog. He was the leader of the Quartet Base for more than 15 years. He started the OGR a guitar quatuor. In 2018, he had to stop all performances for two years because of health issues. This is where the Electron Zero project started. He left the Muzzix collective in 2023. Now he is leading a 3 years project at "La Chambre d'Eay" (Le Favril, France) with the "Reverberations" and "Working sounds" projects. He is also involved in the "Vibrating Shapes" project led by researchers Arthur Paté and Florent Berthaut. He plays in the ensembles "Sentimental Mood", "Inner Dancers", "Blazz" and others. Since 1997, he has played in many venues in France but also abroad : Tokyo, Hué, Amsterdam, Frankfort, Krakow, and others.

Florent Berthaut received a master's degree in 2006 and a Ph.D. in computer science in 2010 at the Université de Bordeaux, France. He is currently Associate Professor at the Université de Lille, France, and head of the MINT team, part of the CRISTAL (UMR 9189 CNRS) laboratory. His research is in the fields of Human-Computer Interaction, Extended Reality and New Interfaces for Musical Expression. In particular, he investigates novel 3D interfaces to enrich the experience of audience members, ensembles and performers.